Zirui Chen, Ph.D. Student

chenzirui@westlake.edu.cn
https://czr-gif.github.io/

Education

2024 – Present — **Ph.D, Westlake University (joint program with Zhejiang University)** Intelligent Unmanned System Laboratory
Mentor: Prof. Shiyu Zhao
Major: Computer science and technology
Tentative Thesis title: *Collective Behavior Clone via Neural Network*

2021 – 2024 — **M.Eng., Beihang University**, The Seventh Devision
Mentor: Prof. Zongyu Zuo
Major: Control Science and Engineering
Thesis title: *Nonsingular Guiding Vector Field for Geometric Path Following Control*.

2017 – 2021 — **B.Eng., Beihang University**, School of Automation Science and Electrical Engineering
Major: Automation Science and Engineering
Thesis title: *Path Following Control for Airship*.

Research Publications

1. Z. Chen, S. Guo, and S. Zhao, "Guiding vector field generation via score-based diffusion model," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, Accepted, Vienna, Austria, Feb. 2026.
2. e. a. Yue Feng and Z. Chen, "Completing the response space: Unifying rejection and multi-interval video grounding via cardinality-rectified grpo," in *2026 IEEE International Conference on Multimedia and Expo (ICME)*, Accepted, 2026.
3. Z. Chen and Z. Zuo, "Non-singular cooperative guiding vector field under a homotopy equivalence transformation," *Automatica*, vol. 171, p. 111 962, Jan. 2025. DOI: 10.1016/j.automatica.2024.111962.
4. Z. Chen, Z. Zuo, G. Wang, P. Li, and Y. Liu, "Discrete data-based frequency-domain closed planar path following method," *IEEE/ASME Transactions on Mechatronics*, pp. 1–12, Jan. 2025. DOI: 10.1109/tmech.2025.3566540.
5. Z. Chen, J. Tang, and Z. Zuo, "A novel prescribed-performance path-following problem for non-holonomic vehicles," *IEEE/CAA Journal of Automatica Sinica*, vol. 11, no. 6, pp. 1476–1484, Jun. 2024. DOI: 10.1109/jas.2024.124311.

Skills

Robotics & Control — Robot learning, Generative model, Nonlinear control design, reinforcement learning, trajectory planning and tracking.

Languages — Native in Mandarin Chinese; fluent in English (IELTS 6.5).

Miscellaneous Experience

Teaching Experience

2021 — **Best Teaching Assistant Award**, Beihang College, Beihang University.

Miscellaneous Experience (continued)

Academic Service

Since 2024 — Reviewer for International Journal of Robotics Research (IJRR), IEEE Robotics and Automation Letters (RAL), Automatica, IEEE Transactions on Industrial Informatics (TII), and Journal of the Astronautical Sciences (JAS).

Awards

2025 — **CAA Outstanding Master's Thesis Award in 2025**, Beijing, China.